Wen-Chieh Chao, MD MS

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Ultrasound-responsive biomaterials • 3D bioprinting

Education

Stanford University 2024 - 2026

Masters of Science - BioEngineering (GPA: 3.93) *Stanford, CA*

- Computational Coursework:
  Machine Learning, Cloud Storage and Computing, Signal Processing, Feedback Control Design
- Biomedical Coursework:
  Quantitative Physiology, Systems Biology, Computational Structural Biology, Tissue Engineering, Biomaterials
- Entrepreneurship Coursework:
  Patent Law and Strategy, Biodesign and Innovation

National Taiwan University 2017 – 2023

Doctor of Medicine - MD, Medicine (GPA: 4.02) *Taipei, Taiwan*

- Clinically trained with full two-year hospital rotations across 30+ departments/specialties
- Provided primary care for 80+ hospitalized patients; assisted in 20+ and observed 150+ surgeries.

Research Experience

Graduate Researcher Oct 2024 – Present

The Skylar-Scott Lab (PI: Mark Skylar-Scott), Stanford - Biomaterials, Liposomes *Stanford, CA*

- Enabled controllable, ultrasound-triggered fibrin crosslinking by engineering a thrombin-loaded, ultrasound-responsive liposome release platform.
- Achieved robust and tunable thrombin release by optimizing membrane mechanics across >100 liposome fabrication conditions.
- Performed iPSC culture experiments to evaluate the effects of a lab-developed engineered growth factor.

Research Assistant Dec 2023 - Jun 2024

Wu Lab (PI: Yu-Wei Wu), IMB Academia Sinica - Neuroanalysis, Neuro-behavior *Taipei, Taiwan*

- Characterized DBS-induced behavioral and neural changes in Parkinson's mouse models by conducting controlled behavioral assays on >20 PD mice.
- Increased reproducibility and throughput of dendritic spine quantification by optimizing DeepD3-based analysis workflows.
- Extracted nonlinear structure in neural datasets by using PCA/UMAP-based dimensionality-reduction analyses.

Research Assistant Apr 2021 - Jun 2021

Brain and Mind Lab(PI: Joshua O. Goh), National Taiwan University - fMRI, VR design *Taipei, Taiwan*

- Identified age-related differences in brain activation by performing full fMRI preprocessing and 2nd-level SPM analysis in MATLAB.
- Enabled controlled testing of spatial navigation behavior by building non-Euclidean virtual environments in Unity.
- Reduced experimental analysis time by 80% by integrating open-source gaze-analytics pipelines into the workflow.

Rotation Internship Oct 2022 - Nov 2022

The Tytgat Institute(PI: Stan van de Graaf), Amsterdam UMC - Receptor screening *Amsterdam, Netherlands*

- Screened bile-salt–responsive GPCR candidates by applying the Presto-Tango reporter platform in HTLA cells.
- Boosted transfection success rate from 50% to 90% by optimizing reagent stoichiometry and workflow conditions.
- Supported project planning by synthesizing bile-salt signaling literature & presenting updates to research team.

Publications & Presentations

Betty Cai, Seungheon Lee, **Wen-Chieh Chao**, Mark Skylar-Scott, Sarah C. Heilshorn *Ultrasound-Sensitive Liposomes for Triggered Thrombin Release.* Poster presentation within program, April 16, 2025.

Technical Skills

Wet Lab & Biomaterials: Liposome fabrication, purification, and controlled cargo release; iPSC culture
Computational & Modeling: MATLAB, Python, fMRI/SPM, neural signal analysis, PCA, UMAP
Instrumentation & Imaging: Ultrasound, confocal/fluorescence microscopy, FIJI, DeepD3